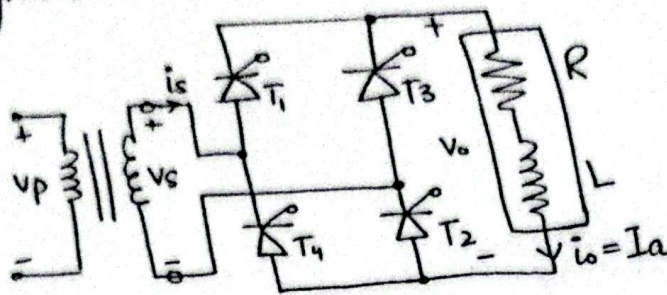


Name : Sata Saij

Roll no : 22L-7734

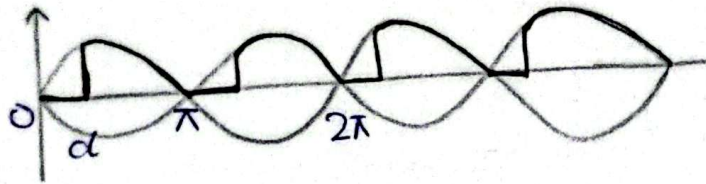
POWER ELECTRONICS - 6A

Assignment 5



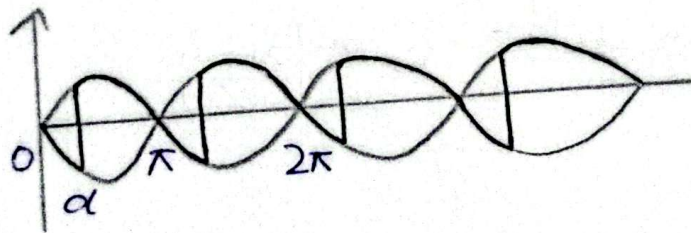
→ if $L=0$ then v_o :

output voltage follows the input voltage from firing angle α to π , and again $\pi+\alpha$ to 2π



→ if $L=\infty$ then v_o :

Full conduction over half-cycle from α to $\pi+\alpha$



3- Determine the dc and rms output voltage if $\alpha = 60^\circ$:

$$\begin{aligned} * V_{dc} &= \frac{2V_m}{\pi} \times \cos \alpha \\ &= \frac{2V_m}{\pi} \cos 60^\circ \\ &= \frac{2V_m}{\pi} \times \frac{1}{2} \\ &= \frac{V_m}{\pi} \end{aligned}$$

$$\begin{aligned} * V_{RMS} &= V_m \sqrt{\frac{1}{\pi} \left(\pi - \alpha + \frac{\sin 2\alpha}{2} \right)} \\ &= V_m \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{3} + \frac{1}{2} \sin \left(\frac{2\pi}{3} \right) \right)} \\ &= V_m \sqrt{\frac{1}{\pi} \left(\frac{2\pi}{3} + \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right) \right)} \\ &= V_m \sqrt{\frac{1}{\pi} \left(\frac{2\pi}{3} \right) + \frac{1}{\pi} \left(\frac{1}{2} \times \frac{\sqrt{3}}{2} \right)} \\ &= V_m \sqrt{\frac{2}{3} + \frac{\sqrt{3}}{4\pi}} \end{aligned}$$

4- $\alpha = 30^\circ$:

$$\begin{aligned} * V_{dc} &= \frac{2V_m}{\pi} \cos 30^\circ = \frac{2V_m}{\pi} \left(\frac{\sqrt{3}}{2} \right) \\ &= \frac{\sqrt{3} V_m}{\pi} \end{aligned}$$

$$\begin{aligned} * V_{RMS} &= V_m \sqrt{\frac{1}{\pi} \left(\pi - \alpha + \frac{\sin 2\alpha}{2} \right)} \\ &= V_m \sqrt{\frac{1}{\pi} \left(\pi - \frac{\pi}{6} + \frac{1}{2} \left(\sin 2 \left(\frac{\pi}{6} \right) \right) \right)} = V_m \sqrt{\frac{1}{\pi} \left(\frac{5\pi}{6} + \frac{1}{2} \times \frac{\sqrt{3}}{2} \right)} \\ &= V_m \sqrt{\frac{1}{\pi} \times \frac{5\pi}{6} + \frac{\sqrt{3}}{4} \times \frac{1}{\pi}} \Rightarrow V_m \sqrt{\frac{5}{6} + \frac{\sqrt{3}}{4\pi}} \end{aligned}$$